Can AI-Human interaction drive sound change? Insights from phonetic accommodation in Hong Kong Cantonese [ŋ-]-[Ø-] onset merger

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Artificial Intelligence(AI)-generated voices are everywhere

Wide application in different fields

- Second language acquisition (e.g., Song et al., 2022)
- Speech therapy tools (e.g., Deka et al., 2025)
- Mental health interactions (e.g., Liu et al., 2025)
- In-home voice assistants (e.g., McLean & Osei-Frimpong, 2019)
- Customer services (e.g., Wang et al., 2023)
- Advertisements (e.g., Wang et al., 2024)
- …

→ 🤖 **Human-AI interaction is the new norm**

Speech is modulated differently in Human-AI interactions

Different phonetic realizations compared to human-human interactions

- *Duration* (e.g., Cohn et al., 2024; Chen et al., 2026)
- *Spectral Qualities* (e.g., Cohn & Zellou, 2021; Chen et al., 2026)
- *F0* (e.g., Bradshaw et al., 2023, Cohn et al., 2022, Hong et al., 2026)
- *Intensity* (e.g., Raveh et al., 2019)
- …

→ **Primarily focus on immediate effects (i.e., within the interactions)**

→ **Could there be any carryover effects (i.e., after the interactions)?**

Could there be any carryover effects after interactions?

Sound Change

- Allows us to systematically examine potential **carryover** effects
- **Before** AI era…
  - Human listeners and speakers primarily contribute changes to their speech community
- **Now we are now in** the AI era…
  - AI-generated speech becomes an additional input to the existing community

→ **Regardless, listeners remain human**

→ **What makes human listeners crucial to sound change?**

We are not perfect listeners 👂 + tend to imitate 🗣️

= Phonetic Accommodation

- Step 1. Listeners: Exposed to a variant
- Step 2. Listener-turned-Speakers: Accommodate 🗣️ the perceived variant (intended or distorted) (Ohala, 2012)
  - Convergence (i.e., phonetic imitation) is the most reported type of accommodation to human voices

- Accommodation contributes to sound change in progress (Garrett & Johnson, 2013)
  - Accommodation ~ a mini-sound change

→ Do listeners also accommodate AI voices?

→ If so, could AI voices influence sound change differently?

Positioning AI voice in sound change

RQ: Does AI have the potential to contribute to sound change?

- Specifically, we examine phonetic accommodation (i.e., a mini-sound change):
- 1. Do accommodation patterns differ **between AI and human voices**?
- 2. Do accommodation patterns remain **after shadowing**?

Positioning AI voice in sound change (Cont.)

🎯 Target feature: Hong Kong Cantonese (HKC): [ŋ-]-[Ø] onset merger

- E.g., 餓 **[ŋɔː^{22}]–[ɔː^{22}]** ‘hungry’
- **Why this sound change?**
  - 1. 🕰️ A bidirectional merger **remains unsettled after 100 years**
    - → Can AI voices **alter its** 📈 **trajectory** 📉 ?
  - 2. The only HKC sound change feature with **a reported loss of the production-perception link** 🔁 (Cheng at al. 2022)
    - → Does 👂 **listener-based accommodation still operate**?

Method - ⭐We trained our own AI speech

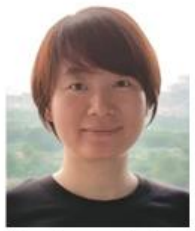
Generated using duration-informed TTS model

- 2 phonetically-trained, native HKC female RA
- 4000 Cantonese sentences each

(Thanks to our engineering team!)



Tan Lee



Yusheng Tian

Stimuli	Meaning	Shadowing Variant Played
我	'I'	[ŋɔː^{23}]
餓	'hungry'	[ŋɔː^{22}]
外	'outside'	[ɔːi^{22}]
哀	'sorrow'	[ɔːi^{55}]
愛	'love'	[ɔːi^{33}]
曖	'ambiguous'	[ɔːi^{25}]

Why customized AI voice?

- Voice manipulation was **NOT** considered previously
- → **Control the variants produced**, esp. in the study of sound change

Method - Exposure to both human and AI voices

Participants (n = 33)

- 21.52 ± 3.76; 17F & 16M
- **Between-subject** factors:
  - Face (Human / Avatar)
  - Model talker identity (RA1 / RA2)
- **Within-subject** factor:
  - Voice (Human / AI)
- **Order** of **voice** is **counterbalanced**

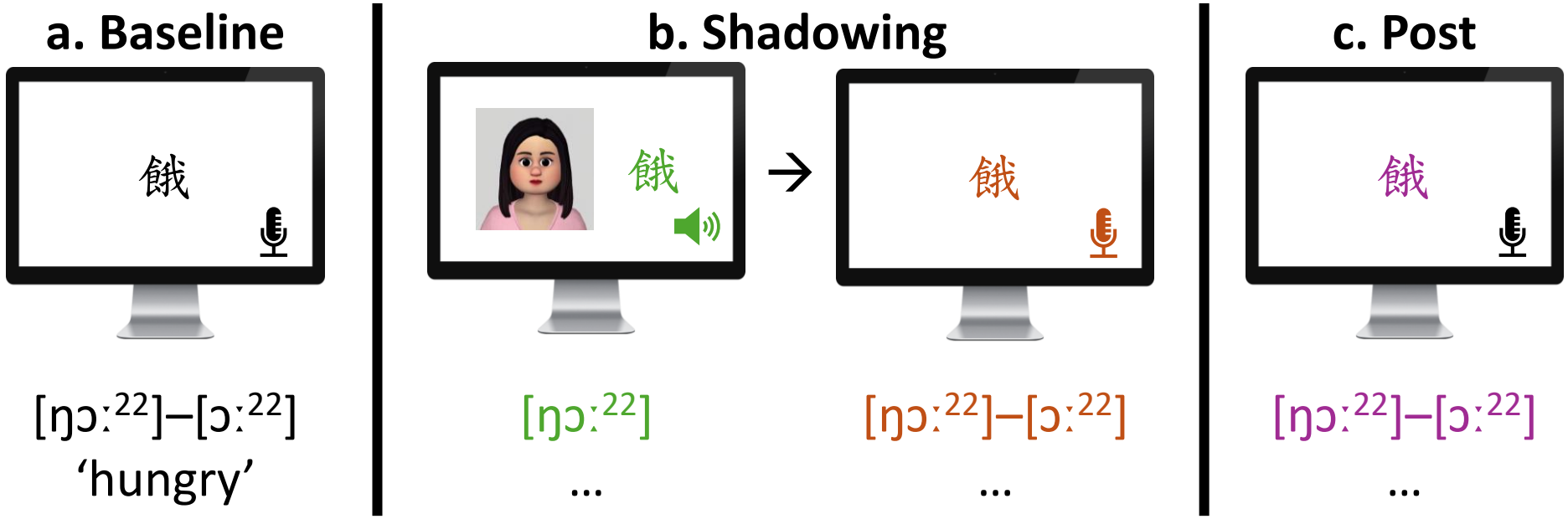
Voice \ Stimuli	餓 [ŋɔː^{22}] ‘hungry’	愛 [Øɔːi^{33}] ‘love’
👩 **Human**		
🤖 **AI-generated**		

→ Closely resemble human voices
→ Subtle perceivable diff. due to synthesis

Method - Assess Accommodation using shadowing task

Shadowing task with a model talker

- 👂 A listener-turned-speaker-based experimental technique



- → Similar mechanism as how a change is initiated and spread (Garrett & Johnson, 2013)

Measurement - Temporal Adjustment = Accommodation

Temporal adjustment as operationalization of **phonetic accommodation**

E.g., Convergence (i.e., become model talker-like)

1. Listen to **shorter** vowel proportions
2. Produce **shorter vowel proportions** when becoming a speaker

🗣️ **Baseline**	ŋ	ɔː
👂 **Listen to**	ŋ	ɔː
🗣️ **During shadowing**	ŋ	ɔː
🗣️ **After shadowing**	ŋ	ɔː

Analysis - Linear Combination

~~Difference-in-difference (DID) approach: classic yet biased~~ (Cohen Priva & Sanker, 2019; MacLeod, 2021)

2 LMMs:

Shadowing / Post-Shadowing ~

Baseline +

ModelTalker*Voice*Face +

$ModelTalkerID_{(RA1/2)}$ +

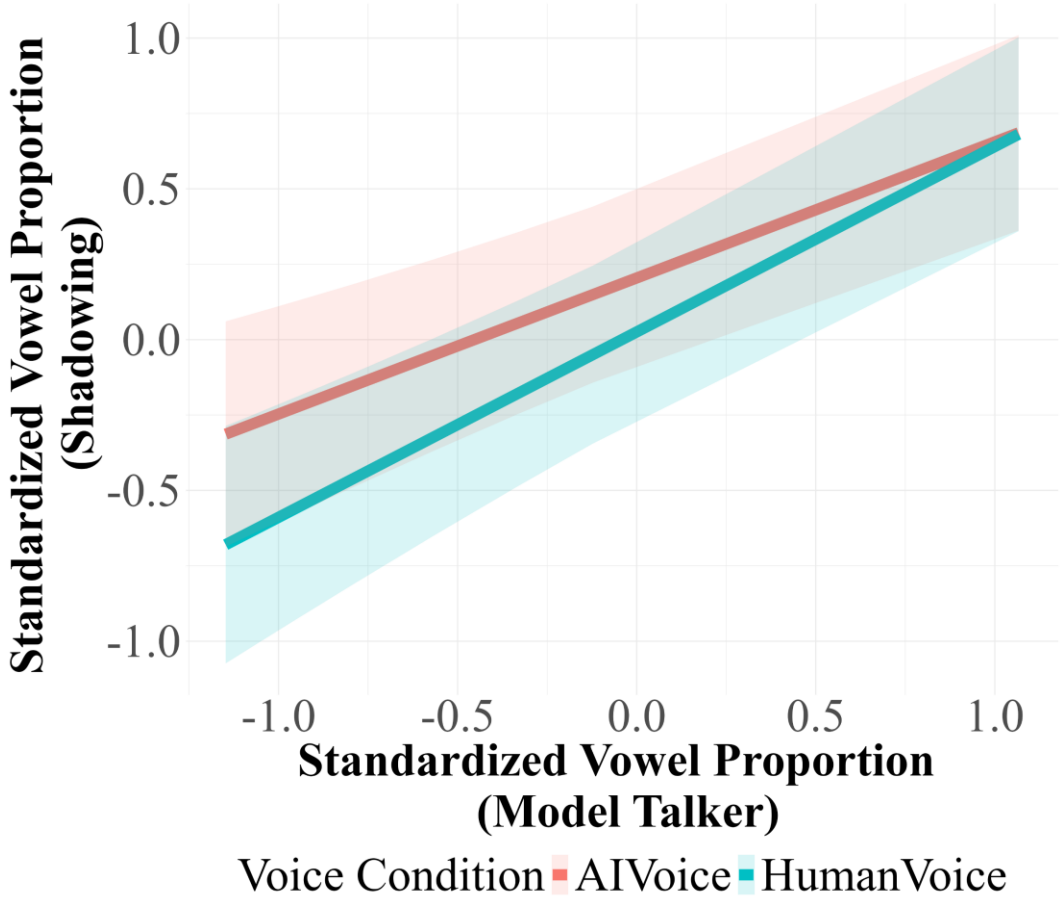
(1|ParticipantID) + (1|Character) + (0 + ModelTalker| ParticipantID)

Diagnostics

✅ Multi-collinearity, Singularity, Homoscedasticity, and Normality of residuals

Results - (1) Convergence found during shadowing

= Longer prop. in the model talker → Longer prop. in participants, VV



Convergence depends on the voice source

- Stronger convergence on the human voices

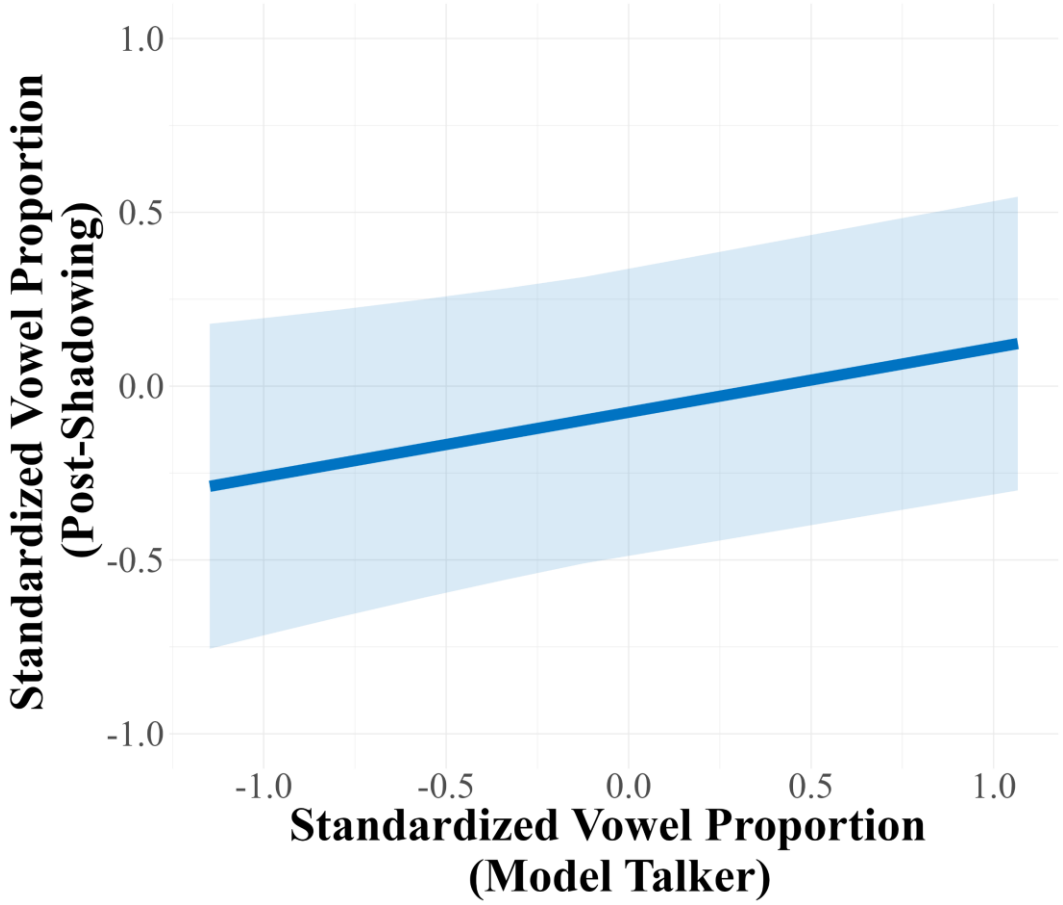
(β = - .25, SE = .07, 95% CI [-0.38, -.11], t(286) = -3.55, p < .001)

- → Participants were **more strongly influenced by human speech** during shadowing
- No other main effects or interactions were sig.

Immediate effects ✅ → Any carryover effects ❓

Results - (2) Convergence retained after shadowing

= Longer prop. in the model talker → Longer prop. in participants, VV



No sig. effects of voice source

(F(1, 290.54) = 0.03, p = .87)

- → No sig. diff. in convergence btw. voices
- → ✅ **Participants' tendency to imitate remained**
- No other main effects or interactions were sig.

Discussion - AI's influence on speech performance

During - Sig. diff. between AI and Human model talkers

- Align with previous finings on Human-AI interaction (i.e., modifications within interactions)

After - No diff. between talkers, yet overall convergence remained

- Comparable effect of AI and human voices over time in affecting speech behaviour

Increasing exposure to AI voices from various sources in recent years

- Multiple short-term carryover accommodations

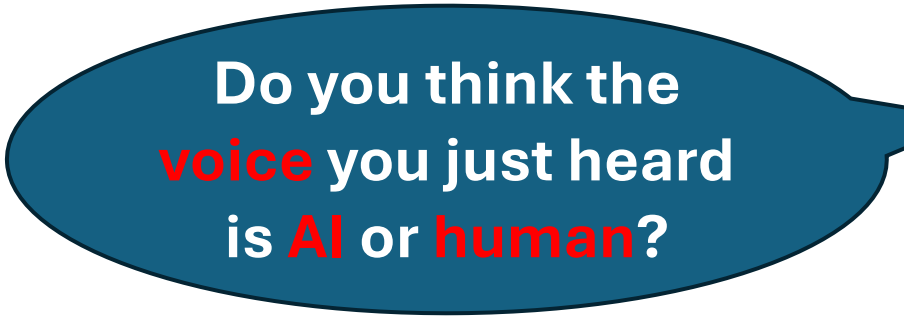
→ A potential impact on sound change

To speech engineer: Variants generated matter!

Discussion - Accommodation is more than just automatic

A socially-mediated behaviour

- Whether a sound is adopted after listening depends on the listener's **social/subjective evaluations of model talkers**
  - E.g., attractiveness, dialect, social status, and other



Face (between) / Voice (within)		Human Face (n = 16)	AI Face (n = 17)
Correctly Identify		8 (50.0%)	5 (29.4%)
Wrongly Identify as	Both AI	3 (18.8%)	4 (23.5%)
	Both Human	5 (31.3%)	8 (47.1%)

- More than 60% participants identified the voice sources **incorrectly**
  - Such awareness could further affect speech performance
  - Yet sample size in each subgroup now is not suitable for any generalization

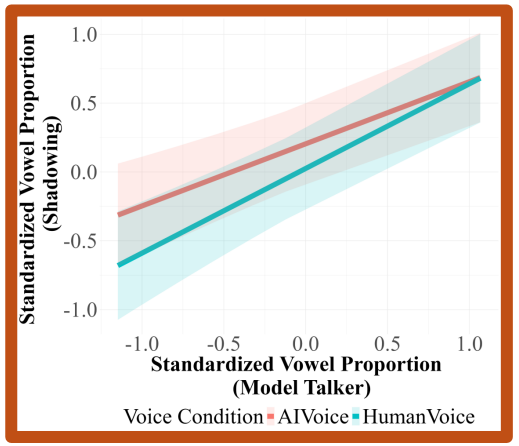
→ More data is needed to address this question

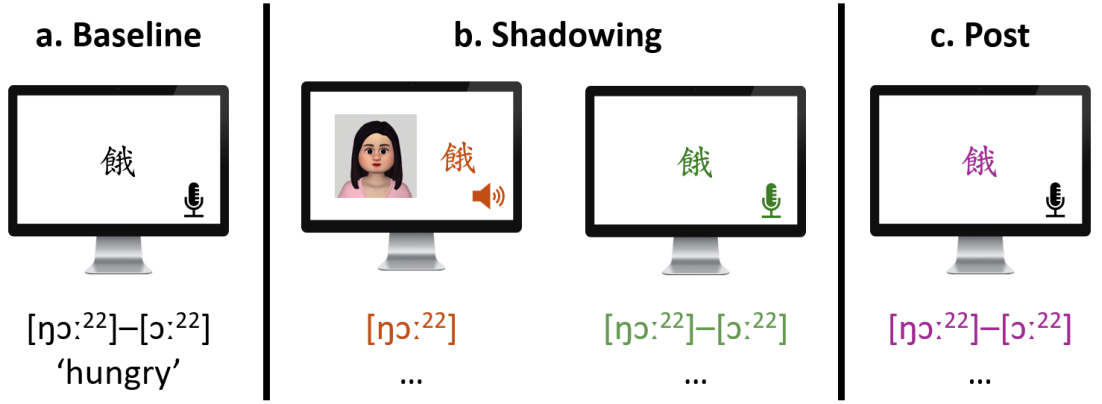
Conclusion - AI has the potential to contribute to sound change?

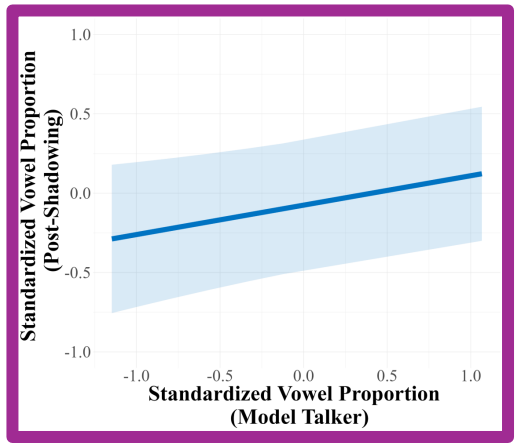
Yes, mini-sound change was found!

- **Convergence (phonetic imitation) during and after exposure**

Stronger influence from human → No sig. diff. btw. (yet imitation retains)





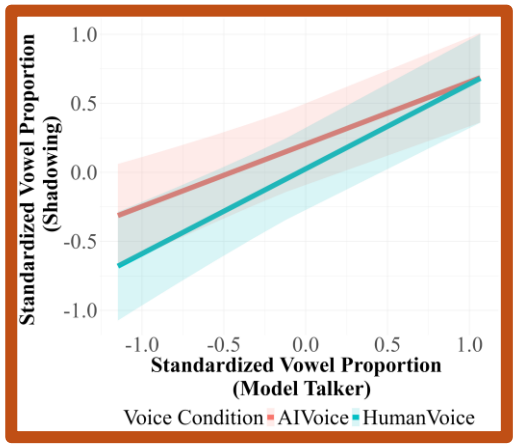


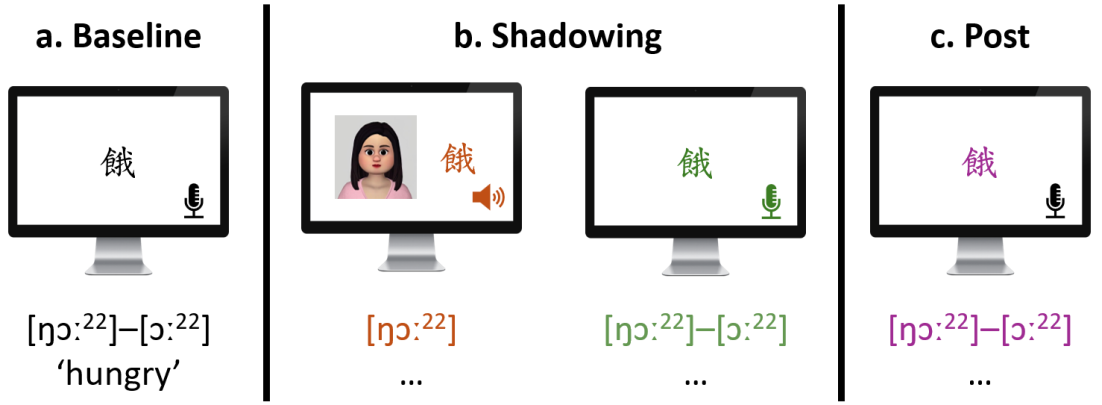
3 Takeaways

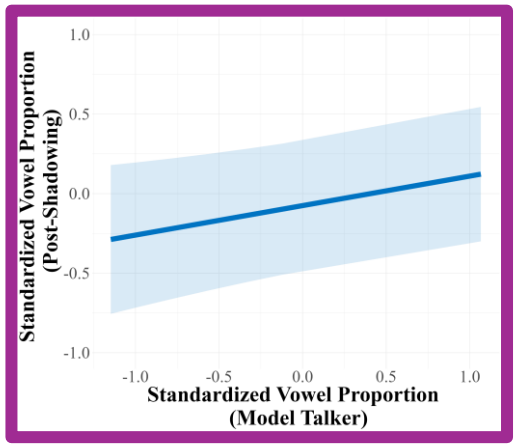
→ Both human and **AI voices** can **influence speakers' speech** **after exposure**

→ **Repeated** **carryover effects** may accumulate to **long-term speech changes**

→ **AI-generated variant** has the potential to **contribute to sound change**







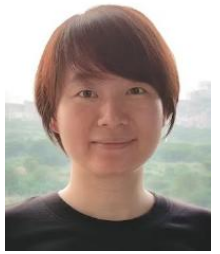
Acknowledgements

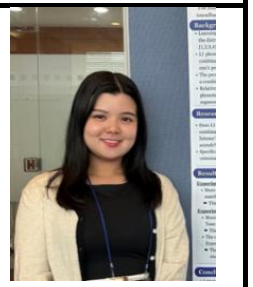
Hong Kong Research Grants Council General Research Fund (Project No. 14612023)
Human vs. AI: Speech register and speech accommodation in human-machine interaction

CUHK Research Committee Postdoctoral Fellowship Scheme 2023–24
Hong Kong RGC Postdoctoral Fellowship

RGC Hong Kong PhD Fellowship Scheme (Reference No.: PF25-16581)

The CUHK Vice-Chancellor HKPFS Scholarship Scheme

Principal Investigator	Collaborators		
 Peggy Mok	 Grace Cao	 Tan Lee	 Yusheng Tian

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 Yitian Hong	 Xinyi Chen	 Xinzi Hou	 Yuanlin Yang	 Ann To
 Karen So	 Manson Wong	 Miko Ng	 Xingni Li	 Gisele Wang

Thank you and comments are welcome!

Slides are available at https://mansonwongcm.github.io/projects.html

